

Aarav Anderson

Embedded Systems & IoT Firmware Engineer • 11 Years Experience • MERIQ Index: 70% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: C/C++, MQTT, I2C/SPI/UART, PCB Debugging, Python, FreeRTOS

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Qualcomm* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core C/C++ service, reducing latency by 55%.
- Led a squad of 7 engineers delivering scalable microservices processing 38M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 40%.

Associate Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2015 - 2023)

- Designed and maintained MQTT data endpoints with 95% SLA uptime.
- Optimized database indexing and caching layers using I2C/SPI/UART saving \$42k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [C/C++, MQTT, I2C/SPI/UART, PCB Debugging]

Engineered end-to-end production solution leveraging C/C++ and MQTT with full test automation.

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [I2C/SPI/UART, PCB Debugging, Python, FreeRTOS]

Architected high-resilience system integrating I2C/SPI/UART with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Robotics and Autonomous Systems

IIT Bombay — 2011 - 2015 (Score: 3.51 / 4.0)